

Relationship of Sedative-Hypnotic Response to Self-Injurious Behavior and Stereotypy by Mentally Retarded Clients

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Groups of clients exhibiting both self-injurious behavior (SIB) and stereotypy, stereotypy alone, SIB alone, or neither of these types of behavior were rated in terms of response to sedative-hypnotic medication. Sixty-eight percent of the clients with both SIB and stereotypy maintained wakefulness and were either resistive, combative, restless, uncooperative, or abusive after treatment with sedative-hypnotics. Thirty-nine percent of the clients with only SIB and 35 percent of the clients with only stereotypy evidenced such "paradoxical" responses to these medications. None of the matched control clients displayed paradoxical responses. This result suggests that an impaired endogenous opiate system characterized by increased levels of β -endorphin may mediate and maintain a type of syndrome of SIB that involves the presence of stereotypy, implying the need for a new diagnostic classification and treatment of these aberrant behavioral disorders.

In a recent study Sandman, Datta, Barron, Hoehler, and Swanson (1983) found that naloxone, an opiate antagonist, was effective in reducing episodes of self-injurious behavior (SIB) and stereotyped self-restraint of two mentally retarded clients. These findings suggested the possibility that a disturbed endogenous opiate system may be involved in mediating and maintaining SIB (i.e., by elevating pain-threshold and/or by the putative addictive characteristics of such behavior). When these findings are coupled with recent reports of normalizing of response to noxious stimulation with naloxone by patients with chronic insensitivity to pain (Dunger, Leonard, Wolff, & Preece, 1980; Yanagida, 1978) and anecdotal observations suggesting that SIB patients who evidence favorable responses to naloxone also displayed "paradoxical" responses to analgesic agents (Sandman et al., 1983), the implications of endorphin involvement in these be-

havioral disorders is supported. It also is plausible that an effective strategy for treating self-injury, and possibly stereotypy, is with putative endogenous opiate antagonists such as naloxone, α -MSH/ACTH fragments (Amir & Amit, 1979; Bertolini, 1979; Sandman & Kastin, 1981), γ -MSH (van Ree, Bohus, Csontos, Gispén, Greven, Nijkamp, Opmeer, de Rotte, van Wimersma Greidanus, Witter, & de Wied, 1981), MSH-inhibiting factors (Kastin, Olson, Ehrensing, Berzas, Schally, & Coy, 1979; Kastin, Nissen, Zadina, Schally, & Ehrensing, 1980) or Melatonin (Datta, Sandman, & Hoeler, in press). Thus, a biochemical substrate or mechanism as an etiological explanation, as well as a practical treatment, for some of the aberrant behavior exhibited by retarded clients may be developed.

Consensus does not exist as to the classification of SIB. The term *self-stimulatory behavior* has been employed to describe both self-injurious and stereotyped behavior or SIB alone (Carr, 1975; Corbett, 1975; Lemke, 1974; Luiselli, Pemberton, & Helfen, 1978), whereas other investigators have classified stereotypy as a type of SIB (Baumeister & Forehand, 1973; Corbett, 1975). Since stereotypy is often found to

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precede SIB and have elements in common with this behavioral pattern, other investigators have suggested that SIB is a consequence or damaging form of stereotypy (De Lissovoy, 1961; Rincover, 1978; Schroeder, Schroeder, Smith & Dalldorf, 1978; Wells, Forehand, Hickey, & Green, 1977); however, in the current study a distinction is drawn between these two behavioral patterns to avoid inferring purpose.

Self-injurious behavior is most frequently reported for individuals of varying ages with clinical diagnoses of infantile autism, schizophrenia, and functional or organic mental retardation, and it is the most prevalent and serious maladaptive behavioral problem exhibited by these institutionalized populations. Self-injurious behavior has been found to occur among 40 percent of institutionalized psychotic children (Green, 1967; Shodell & Reiter, 1968); MacKay, McDonald, and Morrissey (1974) and Maisto, Baumeister, and Maisto (1978) have reported the incidence of SIB for institutionalized retarded children to be as high as 19 and 14 percent, respectively. Although the incidence reports of SIB among mentally retarded individuals vary, the majority of investigators suggest that between 7 to 9 percent are self-injurious (Schroeder et al., 1978; Smeets, 1971; Soule & O'Brien, 1974).

Highly stereotyped behavior is thought to be one of the most pervasive characteristics of autistic and retarded children (Baumeister & Forehand, 1973). In fact, stereotypy is one of the defining characteristics of autism (Rimland, 1964) and is one of the most commonly occurring kinds of behavior observed in the institutionalized mentally retarded population. Incidence reports have indicated that stereotyped behavior is engaged in by at least 66 percent of the institutionalized autistic and retarded populations (Ollendick & Matson, 1978). Thus, the majority of these residents display at least one type of stereotyped behavior (Adams, Tallon, & Stangl, 1980).

The literature suggests that both SIB and stereotypy are of heterogeneous etiology or are multiply determined with no known basic causes or origins. Since it appears

likely that such behavior is motivated by an arousal (stimulation)-controlling or homeostatic condition, there may be common substrates involved in both behavioral patterns.

Presently, there are several treatments used to suppress or eliminate SIB and stereotypy, although these regimens are often criticized because they do not deal with the causes of the disorder except in a superficial way (Carr, 1977). Overall, outcome research indicates that many of these procedures are effective when applied to various intensities of the target behavior, e.g., aversive conditioning in the form of electric shock has been found to modulate successfully moderate and high-intensity or life-threatening behavior (Lovaas & Simmons, 1969; Romanczyk & Goren, 1975). It is not uncommon, however, for an aversive approach to be successful in a highly manipulated environment, with the aberrant behavior reappearing when the client is withdrawn from treatment (Lovaas & Simmons, 1969; Romanczyk & Goren, 1975). Thus, the ameliorative effects of treatment are often short-lived because they are difficult to transfer and do not relate to the cause of the problem.

The present study is an investigation of SIB and stereotyped behavior of mentally retarded individuals. The study was designed to further our understanding of possible etiological factors that may contribute to these kinds of behavior. More specifically, we attempted to investigate the relationship between SIB and stereotypy and the response of affected individuals to sedative-hypnotic medications.

Findings from this study may serve to clarify or substantiate Sandman et al.'s (1983) earlier conclusions. For instance, if we find that persons exhibiting SIBs and stereotypies are "paradoxical" responders to sedative-hypnotic medication, then we may assume that they will respond favorably to treatment or substances that have effects antagonistic to these medications. Further, if our findings are confirmed, then a pharmacological marker may establish a specific biochemical deficit or abnormality related to SIB and stereotypy. Thus, evidence may be provided linking a

physiological substrate or mechanism to these kinds of behavior.

Method

Subjects

Subjects were 100 mentally retarded clients (60 males, 40 females) from the Adolescent Development Program at Fairview Hospital, an institution for retarded individuals in Costa Mesa, California. Clients were divided into four groups: (a) clients exhibiting both SIB and stereotypy; (b) clients with stereotypy, but not SIB; (c) clients with SIB, but not stereotypy; and (4) clients displaying neither of these kinds of behavior.

Clients ranged in age from 9 to 26 years (mean = 20.71, standard deviation [*SD*] = 3.60). Average length of institutionalization was 13.13 years. The most frequent medical diagnoses were mental retardation due to: unknown prenatal influence, other defective fetal development (32 percent); chromosomal abnormality, G group, Down syndrome, trisomy G (9 percent); and trauma or physical agent, perinatal hypoxia (7 percent). The population consisted of 14 percent severely retarded and 86 percent profoundly retarded clients. Their mental ages ranged from 12 to 56 months (mean = 24.78, *SD* = 9.75). There were no significant relationships found between subject group and sex, age, length of institutionalization, level of retardation, and mental age. These data are displayed in Table 1.

Definitions

Self-injurious behavior. This term was used to describe repetitive self-inflicted action that encompasses a number of types of behavior and results in physical harm and tissue damage. Such behavior appears dysfunctional and devoid of any desirable consequences because it "tends to cut off, to remove, to maim, to destroy, to render imperfect some part of the body" (Picker, Poling, & Parker, 1979).

Stereotyped behavior. Stereotyped behavior or stereotypy was used to describe persistent, highly repetitive, invariant motor acts, mannerisms, or sequences that seem to have no apparent function and are not injurious to the individual (Baumeister & Forehand, 1973).

Sedative-hypnotic medication. Sedative-hypnotic drugs (central nervous system [CNS] depressants) are defined by Julien (1978) as any chemical substance that exerts a nonselective general depressant action upon the CNS. These compounds are capable of producing not only behavioral sedation or sleep but also behavioral alterations (e.g., relief from anxiety and analgesia).

Since the sedative-hypnotic compounds are general depressants, their effect appears to result from an action on two arousal centers within the brain: (a) the ascending reticular activating system (the ARAS) and (b) the diffuse thalamic projection system. These two centers are located

TABLE 1
SUBJECT CHARACTERISTICS BY GROUP AND SEX

Variable	Group 1		Group 2		Group 3		Group 4		Total
	Male	Female	Male	Female	Male	Female	Male	Female	
N	10	12	24	16	13	5	11	9	100
Age ^a	21.5	24.0	24.1	18.8	21.2	18.4	19.0	19.1	20.7
Length of institutionalization	14.6	12.0	15.5	11.4	11.21	12.6	12.7	12.0	13.1
Level of retardation (%)									
Severe	20.0	8.3	20.8	12.5	23.0	0.0	18.2	11.1	14.0
Profound	80.0	91.7	79.2	87.5	77.0	100.0	81.8	88.9	86.0
Mental age ^b	24.1	22.6	27.8	22.6	27.8	18.3	26.0	22.1	24.78

Note. Group 1 = SIB and stereotypy, Group 2 = stereotypy, no SIB, Group 3 = SIB, no stereotypy, Group 4 = neither behavior.

^a Mean years.

^b Mean months.

within the brainstem and midbrain and are important for the maintenance of behavioral arousal. Julien (1978) hypothesized that the behavioral depression induced by the sedative-hypnotic drugs result from depression of a synaptic transmission processes within these centers.

"Paradoxical" responses. Clients were classified as exhibiting a "paradoxical" response if large doses (those exceeding the normal range) of sedative-hypnotic medication were reported as maintaining wakefulness and producing either resistive, combative, restless, uncooperative, or abusive behavior.

Normal response. Clients were classified as exhibiting a normal response if the prescribed sedative-hypnotic medication satisfactorily/effectively sedated or anesthetized them.

Design

Two designs were used to analyze the data of this study. The primary design was a 2 (presence or absence of SIB) $\times$ 2 (presence or absence of stereotypy). The number of clients exhibiting "paradoxical" responses to sedative-hypnotics were assessed by chi-square analysis within this design. A second design was constructed post hoc. In this design whether or not a client responded "paradoxically" was included as a factor in the analysis in order to examine the impact of contributing variables.

Procedure

One-hundred mentally retarded clients were assigned to one of the defined experimental groups. Demographic and descriptive information were obtained using a behavioral survey checklist/questionnaire. Each clients' medical records were evaluated to determine whether they exhibited normal or "paradoxical" responses to sedative-hypnotic medication. The resulting data were analyzed in accordance with the method described later.

Instrumentation

Self-injurious and stereotypy behavior survey checklist/questionnaire. The behavior checklist/questionnaire consisted of 44 items that were adapted from a recently revised Fairview Problem Behavior Checklist. There were 22 items that represented each of the behavioral patterns under investigation.¹

The checklist/questionnaire consisted of three major categories: (a) demographic and background information, which included name, Fairview identification number, residence number, chronological age, sex, admitting date, diagnosis, and mental age (MA); (b) history of SIB, which included topology, frequency, severity, management response (treatment), and antecedent or provoking events ratings; and (c) history of stereotypy or rhythmic movements, which included topology, frequency, severity, management response (treatment), and antecedent or provoking response ratings.

Response to sedative-hypnotics. The 1978 through 1981 pharmacological profiles of each subject were evaluated to determine the name and strength of the sedative-hypnotic medication that they were prescribed when receiving routine annual clinical examinations. Behavioral observations of the subject's most recent electroencephalogram recordings were also noted. Clients were classified as "paradoxical" or normal responders by raters unaware of the hypothesis or intent of this study. The body weight and daily medication of each client were also noted.

Method of Data Analysis

Chi-square tests were performed to assess whether an association existed between experimental group, response to medication, and type of sedating medication prescribed. Further chi-square tests provided information as to whether an as-

¹ A checklist of the self-injurious and stereotyped behavior survey items are available upon request from the authors.

sociation existed between experimental group, demographic variables, topology of behavior, management response, and antecedents.

In the second, post hoc analysis, the independent variables evaluated were SIB, stereotypy, and response to sedating medication. The dependent variables were: (a) presence of sensory handicap, (b) use of seizure medication, (c) frequency of occurrence of the behavior scored, (d) severity of the behavior scored, (e) amount of sedating medication prescribed, and (f) weight. To determine the effects of these independent variables, we conducted separate factorial analyses of variance for each dependent variable.

Results

The results of the chi-square analysis suggested that a higher proportion of clients who exhibited SIB and/or stereotypy also displayed "paradoxical" responses to sedatives as compared to those who do not exhibit either type of behavior. In Table 2, the frequency and percentage of clients having a normal or "paradoxical" response to sedative-hypnotic medication is summarized for each of the experimental groups. Analysis of these data resulted in a significant overall association between behavioral response to sedatives and presence or absence of SIB and/or stereotypy, $\chi^2 (3) = 21.21, p < .001$. Thus, these data dramatically suggested that clients with SIB and/or stereotypy have unusual responses to medications commonly used for sedation.

A Ryan's specific comparisons analysis

indicated that the statistical difference among the groups is accounted for most by the group displaying both SIB and stereotypy. This group differed significantly, $p < .05$, from all of the other groups (SIB: $\chi^2 (1) = 7.28$; stereotypy: $\chi^2 (1) = 7.48$; control; $\chi^2 (1) = 9.80$. The differences among the remaining three groups did not differ significantly in response to sedative-hypnotic medication; therefore, clients who engage in both SIB and stereotypy are most likely to have "paradoxical" responses to medication than those who exhibit only one or neither of these types of behavior.

In order to determine other effects of SIB and stereotypy, we conducted a two-way analysis of variance on the following variables: (a) presence of sensory handicap(s), (b) use of seizure medication, (c) frequency of SIB and/or stereotypy, (d) severity of SIB and/or stereotypy, (e) amount of chloral hydrate prescribed, (f) amount of seconal prescribed, (g) amount of thorazine prescribed, and (h) weight. Results suggested that in the population sampled, there were statistically significant effects for the experimental groups. Table 3 shows the F ratios for the various analyses.

There were two significant main effects for stereotypy. The mean for stereotypy differed significantly from the other means on sensory handicap, $F (1, 96) = 6.71; p < .05$, and on frequency of behavior, $F (1, 96) = 1151.79, p < .001$. Thus, significantly more hearing and/or visual impairments appeared among the clients exhibiting stereotypy and the stereotypic behavior that they emitted occurred at a higher rate. The mean for SIB differed significantly

TABLE 2
RESPONSES TO SEDATIVE-HYPNOTIC MEDICATION FOR EACH GROUP

Response to sedative- hypnotic medication	Group 1		Group 2		Group 3		Group 4		Frequency marginal total
	Fre.	% of group total	Fre.	% of group total	Fre.	% of group total	Fre.	% of group total	
Normal	7	0.32	26	0.65	11	0.61	20	1.00	64
Paradoxical	15	0.68	14	0.35	7	0.39	0	0.00	36
Marginal total	22	1.00	40	1.00	18	1.00	20	1.00	100

Note. Group 1 = SIB and stereotypy, Group 2 = stereotypy, no SIB, Group 3 = SIB, no stereotypy, Group 4 = neither behavior.

TABLE 3
F RATIOS (1, 96 *df*) FOR THE DEPENDENT VARIABLES BETWEEN SIB (A) and STEREOTYPY (B)

Variable	Source		
	A	B	A × B
Sensory handicap	0.23	6.71*	0.23
Seizure medication	0.00	0.11	0.03
Frequency of SIB and/or stereotypy	700.92	1151.79**	1000.54**
Severity of SIB and/or stereotypy	362.79**	12.59*	100.47**
Amount of chloral hydrate	0.68	0.26	0.35
Amount of seconal	0.02	2.21	1.55
Amount of thorazine	2.38	0.84	1.42
Weight	0.00	0.01	1.55

* $p < .05$.

** $p < .001$.

from the other means on severity of behavior, $F(1, 96) = 363.79$, $p < .001$, indicating that SIB cases caused more self-directed injury to themselves than did others. The findings imply that stereotypy is significantly related to severity of behavior. The latter two findings may be thought to be definitional and part of the criteria used for group selection.

The effect of "paradoxical" response to sedative-hypnotic medication was assessed by a one-way analysis of variance. Twelve dependent variables were analyzed post hoc: age, length of institutionalization, MA,

sex, presence of sensory handicap(s), use of seizure medication, frequency of SIB and/or stereotypy, severity of SIB and/or stereotypy, amount of chloral hydrate prescribed, amount of seconal prescribed, amount of thorazine prescribed, and weight. Results suggested that there were no statistically significant effects for response to sedative-hypnotic medication. The means, *SD*s, and *F* ratios for the dependent variables between the two levels of response to sedative-hypnotic medication are presented in Table 4. Further, there was not a significant association between type

TABLE 4
MEANS, *SD*s, AND *F* RATIOS FOR THE DEPENDENT VARIABLES BETWEEN THE TWO LEVELS OF BEHAVIORAL RESPONSE TO SEDATIVE-HYPNOTICS

Variable	Behavioral response to sedative-hypnotics				<i>F</i> ^a
	Normal		"Paradoxical"		
	Mean	<i>SD</i>	Mean	<i>SD</i>	
Age	20.79	3.55	20.56	3.72	0.75
Length of institutionaliza- tion	13.22	5.00	12.96	5.71	0.06
Mental age	25.29	9.74	23.92	9.84	0.46
Sex	1.38	0.49	1.43	0.50	0.25
Sensory handicap	1.92	0.27	1.89	0.32	0.23
Seizure medication	1.66	0.48	1.70	0.46	0.14
Frequency of SIB and/or stereotypy	5.48	0.57	5.46	0.51	0.02
Severity of SIB and/or stereotypy	1.44	0.64	1.55	0.63	0.62
Amount of chloral hydrate	1706.35	811.38	1837.84	950.54	0.54
Amount of seconal	36.51	84.82	67.57	115.60	2.38
Amount of thorazine	6.31	15.54	7.43	14.80	0.13
Weight	112.74	25.37	112.43	27.83	0.00

^a $df = 1, 98$.

of sedating medication and behavioral response.

Overall, we assumed from the results of this study that "paradoxical" response to medication commonly used for sedation was related primarily to presence of both SIB and stereotypy.

Discussion

The results of this study indicate that a higher proportion of clients who displayed SIB and stereotypy had unusual or "paradoxical" responses to sedative hypnotic medication when compared to clients who did not exhibit either of these kinds of behavior. Specific-comparisons revealed that among clients who displayed both SIB and stereotypy there was a significantly higher proportion of "paradoxical" responders than among those who exhibited SIB alone, stereotypy alone, or neither of these types of behavior.

Analysis of possible antecedents or correlative behavior indicated that sensory handicap and frequency of behavior was related to stereotypy, and severity of behavioral disturbance was related significantly to SIB. Both of these findings are consistent with results of previous studies and contribute to the definition of the respective disturbances. Of the variables examined, however, except for SIB and stereotypy, there were no other relationships among demographic factors, antecedent behavior, and "paradoxical" response. These findings suggest that "paradoxical" response to sedative-hypnotics may describe a unique subgroup of clients with SIB and stereotypy.

The current study was prompted by the observation (Sandman et al., 1983) that two clients with "intractable" SIB had a history of "paradoxical" response to sedative-hypnotics. Further, SIB by these clients was greatly attenuated with treatment of naloxone. Sandman et al. (1983) suggested that the endorphin system may sustain SIB and stereotypy either by enhancing the perception of pain (Buchsbaum, Davis, & Bunney, 1977; Grevert & Goldstein, 1977) or stimulating endogenous opiate production in order to provide pain relief. De Lis-

sovoy (1961) and Harkness and Wagner's findings (cited in Carr, 1977) that SIB was produced in response to painful middle ear infection (i.e., otitis media) may support this contention. Both hypotheses (i.e., SIB is caused by an apparent absence of pain reaction [heightened endorphin levels] and SIB is motivated by its analgesic properties [endorphin production]) are compatible; however, these hypotheses do not answer the question as to why SIB would persist habitually. An answer may be found in the literature suggesting that endorphins produce tolerance and dependence, the two components of drug addiction (Leybin, Pinsky, & LeBella, 1976; Ungar, Ungar, Malin, & Sarantakis, 1977; Wei & Loh, 1976). Thus, SIB may be maintained because it supplies the "fix" for an endogenous opiate addiction. The findings from the Seiden and Dykstra study (cited in Picker et al., 1979) may support this hypothesis. They found that when narcotics (exogenous opiates) were withdrawn from physically dependent animals, SIB was produced; therefore, such behavior may become rewarding or pleasurable because they prevent the symptoms of withdrawal by stimulating the production of endogenous opiates (endorphins) to replace the exogenous opiate concentrations. The finding that the endorphins have positive reinforcing properties (Belluzzi & Stein, 1977) is also a corollary to the SIB/ endorphin addiction hypothesis.

Although this hypothesis may be interpreted as indicating that SIB is mediated and maintained by an impaired endogenous opiate system, the data are complicated and difficult to interpret because some self-injurers show normal pain reactions (Carr, 1977), not all head-bangers are found to have otitis (Carr, 1977), and not all animals experiencing narcotic withdrawal exhibit SIB (Picker et al., 1979). Taken as a whole, the evidence reviewed indicates that some type of SIB may be indexed by "paradoxical" response to sedative-hypnotics and a disruption of the endogenous opiate system. Thus, such an impairment may not contribute to SIB in general but, rather, it may be related to a certain type of syndrome of SIB. When the data obtained in

the Sandman et al. (1983) investigation is combined with evidence suggesting that endorphins mediate increased activity (Belluzi & Stein, 1977) and the major findings reported in the present study, we suggest that an impaired endogenous opiate system mediates and maintains a syndrome of SIB, which also involves the presence of stereotypy.

Several investigators have suggested that an excess of endorphins mediates increased activity. For instance, Belluzi and Stein (1977) have found that an excess of endorphins may mediate the behavioral state of mania. Since Davis, Buchsbaum, and Bunney (1979) reported that manics are more pain tolerant than are control subjects, one may infer that mania or states characterized by hyperactivity are associated with an elevation of endogenous opiates.

In the present study, the SIB and stereotypy group is characterized by the highest frequency of activity when compared to the other groups. Since the self-injurious and stereotypic behavior that these clients emit is no different in frequency and severity of the SIB only and stereotypy only groups, one may conclude that they are more active. These findings are supported by investigators who have found that adrenergic stimulation (produced by injections of amphetamines and caffeine) is associated with the appearance of SIB and stereotypy (Lycaki, Josef, & Munetz, 1979). Since increased activity and higher pain thresholds are both considered behavioral responses of the endorphins (Coy & Kastin, 1980), one may imply that the hyperactivity exhibited by the SIB and stereotypy group is a critical component of the syndrome.

The major findings from the present study may lend further support to the hypothesis that an impaired endogenous opiate system is an etiological factor for some clients who exhibit both SIB and stereotypy. Since most, but not all, clients with both types of behavior were found to display "paradoxical" responses (e.g., wakefulness combined with exacerbation of SIB and stereotypy, resistiveness, etc., to sedative-hypnotic medications, i.e., bar-

bituates and CNS depressants), a finding also reported in the Sandman et al. (1983) investigation, we speculate that the endogenous opiate system is involved.

The extrapolation from "paradoxical" response to sedative-hypnotic compounds to an impaired endogenous opiate system can be logically supported by extending Hughes and Synder's model (Hughes, 1976), which has implicated the endorphins in opiate dependence, withdrawal, and addiction. Since marked likenesses between the sedative-hypnotics and opiates have been found with regard to their behavioral effects, site and mechanism of action (i.e., their specificity for the neurotransmitters serotonin [5-HT] and norepinephrine [NE], whose cell bodies are located in the brainstem) and adverse reactions, there is reason to believe that this model may likewise be applied to sedative-hypnotic agents.

Hughes and Synder's model (Hughes, 1976) suggests that the endorphins act pre-synaptically to control the release and, thus, inhibit the neurotransmitters of the brain. When exogenous opiate agonists (i.e., narcotic drugs) are present and a state of analgesia has been produced, Hughes hypothesized that inhibitory control passes from the endogenous opiates (i.e., endorphins) to the exogenous opiates. Since exogenous opiates produce tolerance and increasing amounts of them are needed to maintain the "inhibitory mechanisms at a level of activity . . . sufficient to prevent signs of withdrawal" (Kosterlitz & Hughes, 1975, p. 95), the CNS becomes entirely dependent upon them. Tolerance is thought to result as a consequence at either the pre-synaptic opiate receptor site or the post-synaptic transmitter receptor site or to have a modulatory effect on the nerve terminal in which both endorphin and transmitter are present (Kosterlitz & Hughes, 1975). Hughes (1976) suggested that when opiate narcotics are withdrawn, the receptors remain in a state of tolerance and the endorphins are unable to perform their inhibitory function. This is related to their relatively short-action due to rapid enzymatic degradation, which is not enough to overcome desensitization. Thus, a sudden loss of in-

hibitory control leads to overactivity of various excitatory systems, and withdrawal symptoms occur.

The implication that endorphins play an essential role in opiate withdrawal (Hughes, 1976) is thought to support the major finding of the present study. We hypothesize that the "paradoxical" response to sedative-hypnotics that was reported in the majority of clients with both SIB and stereotypy is a type of withdrawal response. This response may only be evident after sufficient amounts of sedative-hypnotic compounds have been administered to clients who normally have elevated blood levels of β -endorphins. Although withdrawal symptoms are not usually a result of single doses of sedative medications, those who have become physically dependent on a certain endorphin level that has not been inhibited due to sedation may respond with signs of overactivity or withdrawal, which we have labeled "paradoxical response." Such responding may therefore be exhibited by endorphin-dependent individuals because the sedative-hypnotic compound is like a low dose to their endorphin system. This low-dose compound may be thought to induce behavioral excitement because the inhibitory neurons within the brain are depressed, which causes a state of "disinhibition" until endorphin receptors can gain restoration of normal endorphin synthesis.

The implication of this hypothesis is that tolerance or dependence on high levels of the naturally occurring endorphins is possible. Although this has not been proven, prolonged exposure to exogenously administered endorphins has been shown to induce addiction (Wei & Loh, 1976).

Evidence has been presented that supports the hypothesis that an impaired endogenous opiate system may mediate and maintain a type of syndrome of SIB that involves the presence of stereotypy. The finding that clients who exhibit both SIB and stereotypy have paradoxical responses to sedative-hypnotic compounds may indicate that they have excessive levels of β -endorphin. This assumes that a biological substrate or mechanism contributes to this

behavioral syndrome. The implication of endorphin involvement suggests that their modulatory influence must be strongly controlled by feedback mechanisms. Feedback may be obtained through the reduction of the availability of endorphins, a reduction of the receptor sensitivity to available endorphins, and production of physiological antagonists. An effective strategy for treating clients with both SIB and stereotypy may be endogenous opiate antagonists, MSH/ACTH fragments or MSH-inhibiting factors. Under these circumstances, the level of β -endorphin (in blood or cerebrospinal fluid) may provide a critical biochemical marker for treatment with various drugs.

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